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Modification in active power-frequency loop of virtual synchronous generator to improve the transient stability

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ABSTRACT

Virtual synchronous generator (VSG) control scheme has received much attention from researchers as the introduction of rotational inertia to inverters. However, maintaining power-angle stability during the transient state of low inertia microgrids consisting of VSG based distributed generation units is a critical challenge. According to the equal-area criterion, the accelerating and decelerating area play an important role in power system stabilisation. Therefore, a new control strategy based on the equal-area criterion is proposed in this paper. According to the proposed control technique, the auxiliary inertia is incorporated into the governor unit of VSG with an additional power loop. The auxiliary inertia term minimises the difference between the governor unit of VSG and the governor unit of the conventional synchronous generator (SG). In contrast, the additional power loop supplies the auxiliary power during the transient process only when a significant fault occurs. Moreover, to enhance the transient stability, auxiliary inertia helps to improve the convergence time and dynamic characteristics of VSG, and additional power helps to shorten the acceleration area and widen the deceleration area. MATLAB/Simulink based simulations analysis and results validate the effectiveness of proposed control.

1. Introduction

Nowadays, a large number of renewable energy resource-based distributed generator units are integrating into conventional power systems by electronic convertors. These renewable energy resource-based distributed generators have very low, or in some cases, no inertia and are susceptible to significant power fluctuations [1–3]. Moreover, most of the conventional power systems are provided by synchronous generators (SGs), and the SG rotor has rotational inertia and damping, which fluctuate under the load disturbance and affects the stability of the power system. The kinetic energy of the SGs' rotor is released to suppress the change of power and frequency during the disturbance. However, renewable energy resource-based distributed generator has no or low inertia and damping, which is critical to stabilising the power and frequency of the power system [4,5]. Therefore, to suppress the instability of DGs, a virtual synchronous generator (VSG) control technique is proposed, which mimics the inertial and damping characteristics of SG.

The early concept of VSG is proposed in various literature [6–9]. The basic control of VSG utilises the electronic inverter to mimic the operational characteristics of SG. Moreover, the operational characteristics of SG and VSG can also be compared, such as Q-δ, P-f [10]. Consequently, in [11,12], the proposed VSG scheme mimics the swing equation of SG, while the proposed technique in [13] reflects the proper operating characteristics of the synchronous machine. However, current-controlled VSG is equivalent to the current source, so it is challenging to provide voltage and frequency support for the system. Therefore, to overcome the deficiencies of current-controlled VSG, a voltage-controlled VSG technique is proposed in [14].

The essence of voltage-controlled VSG techniques is to simulate the rotor inertia and system frequency modulation characteristics of SG in frequency control, to improve the frequency stability of the system. The power controller and voltage frequency controller make VSG perform functions of power control and frequency modulation [15–17].

Furthermore, various techniques of VSG are presented in [17–21], which evolves the model and control to mimic the different control

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dynamics of SG. These models and control techniques achieve the working equivalence and self-synchronisation of SG without a phase-locked loop (PLL). However, VSG control is still needed to be mature enough in theoretical and practical application in comparison to an SG [22]. Considering the transient stability of VSG based power systems, three critical techniques are used to analyse the stability issue named Lyapunov function technique, equal area criterion, and centre of inertia technique [23–25]. In [23], Lyapunov's function method based transient stability analysis is carried out for a grid connected VSG. The deteriorative effect of reactive power control loop on transient angle stability is analysed, and voltage variation is incorporated into an approximate Lyapunov's direct method. According to the proposed method, the inverter internal voltage is treated as a parameter rather than a state variable.

In [25,26], the centre of inertia technique is used to analyse the transient stability of VSG. In [26] researcher proposed the modification in the governor block to improve the transient stability of the VSG and analyse it according to the centre of inertia reference frame. Moreover, transient stability analysis is performed for a small test system as well as for an extensive system, and simulation analysis shows that the proposed technique improves the transient stability of VSG significantly. However, further research is required for optimal integration location of VSG in the extensive power system. In [27], the process of frequency oscillation during disturbance is theoretically analysed using power-angle curve and equal area criterion method, with the conclusion that the deviation and changing rate of angular velocity can be changed by inertia time constant of VSG.Additionally, in [28,29], authors use the virtual mechanical power to analyse and evaluate the impact on active and reactive power of invertor as well as power grid stability.

However, to improve the transient stability of VSG a new control technique is proposed in this paper. The proposed modification is incorporated into the conventional VSG by incorporating an auxiliary inertia term in the governor unit with an additional power loop. The contributions of the proposed modification are following.1. The auxiliary inertia term minimises the difference between governor unit of VSG and governor unit of the SG.2. The additional power loop supplies the auxiliary power during the transient process only when a significant fault occurs.3. The additional power term helps to reduce the acceleration area of VSG and increase the deceleration area and improve the transient power-angle stability.

The remainder of this paper is organised as follows; Section 2 describes the VSG control structure. Section 3 explains the power-angle characteristics, and the proposed modification in VSG control is described in Section 4. Whereas, simulation results are presented in Section 5, and Section 6 summarises the current study.

2. VSG control structure

A basic and pretty simple structure of conventional VSG is shown in Fig. 1, and it can be observed that renewable energy resource-based distributed generator consist a VSG which possesses an energy storage device, DC/AC converter, a filter circuit, governor unit and connected to a grid. If the power of the distributed generator and energy storage system is assumed as the input torque of the prime mover, while DC/AC converter is assumed as the electromechanical energy transformer between stator and rotor. Then, the fundamental component of midpoint voltage is representing the electromotive force of the VSG. Resistance and inductance of filter unit representing the stator winding impedance.

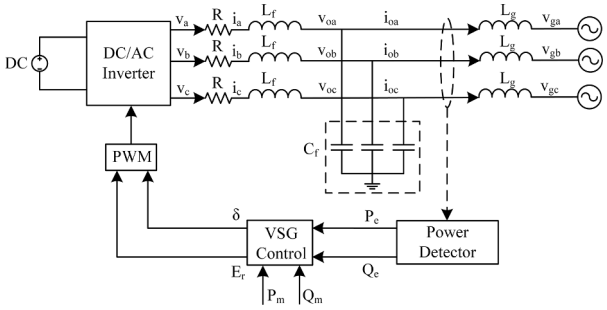


Fig. 1. VSG Basic structure.

According to Fig. 1, v_{abc} and i_{abc} are the AC side voltage, and current of the inverter, respectively. v_{oabc} and i_{oabc} are the LC filtered voltage, and current, respectively. E_r and δ are the internal potential amplitude and phase angle of VSG. v_{gabc} is the grid voltage, and R, L_f and C_f are the filter resistance, inductance, and capacitance, respectively. L_g is the gridline inductance, and P_e, P_m and Q_e, Q_m are the active power and reactive power output by VSG.

Various VSG control strategies provide active and reactive power control and voltage and frequency stability. Generally, VSG control strategies have been developed to enable an electronic convertor to mimic the characteristics of an SG. The simple swing equation of SG is used as a core part of VSG, and it is stated as

$$\begin{aligned} J\frac{d\omega}{dt} &= T_m - T_e - D(\omega - \omega_r) \\ \frac{d\delta}{dt} &= \omega \end{aligned} \tag{1}$$

where J is the virtual inertia, D is the damping coefficient, ω is the virtual angular frequency, ω_r is rated angular frequency, T_m is the mechanical torque, T_e is electromagnetic torque, and δ is the power angle. The input mechanical torque can be calculated as:

$$T_m = \frac{P_m - P_{ref} - k_p(\omega - \omega_r)}{\omega_r} \tag{2}$$

where P_m is the inverter input power, P_e is the output electrical power of the inverter. To mimic the electromagnetic characteristics of SG, for VSG, the stator electrical equation of the synchronous generator is generally simulated without considering the electromagnetic relationship between stator and rotor. According to Fig. 1, it can be expressed as:

$$L_f\frac{di_{abc}}{dt} = e_{abc} - v_{abc} - Ri_{abc} \tag{3}$$

Furthermore, the control structure of the VSG active power-frequency loop with a simple governor is illustrated in Fig. 2. In order to calculate the reference phase and frequency of modulating signal, the active power-frequency loop of VSG mimics the primary frequency regulation, damping and inertia of SG. Whereas, modulating signal amplitude is calculated by a reactive power-voltage loop which mimics the voltage regulation of SG. The mathematical equation for active power-frequency loop of VSG including the simple governor is described as:

$$J\frac{d\omega}{dt} = \frac{P_m}{\omega} - \frac{P_e}{\omega} - D(\omega - \omega_r) \tag{4}$$

3. Power-angle characteristics

The equal are criterion is a common tool to analyse the transient

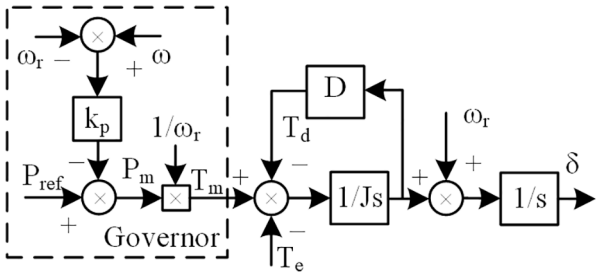


Fig. 2. Control diagram of VSG.

stability of SG based power system. Therefore, the same method can be utilised to analyse the transient stability of the VSG based power system. The power output of a grid connected SG, and ideal VSG can be stated as:

$$P_e = \frac{EV}{X_\Sigma}\sin\delta = P_u \sin\delta \tag{5}$$

where, X_Σ is the sum of filter reactance and line reactance, P_u is the amplitude of VSG output power, and E and V are the input and output voltage of VSG. Subsequently, during the fault in the power system, VSG goes in the unsaturated region due to the high resistance of the line. Therefore, the power-angle characteristic curve of the ideal VSG is the same as of SG, as shown in Fig. 3.

Considering Fig. 3 shows the acceleration area and deceleration area, respectively. While, P_1, P_2, and P_3 is pre-fault or stable state electrical power, post-fault or marginally stable electrical power, and the electrical power during the fault, respectively. While δ_0 is the initial angle at stable state, δ_c is clearing angle at fault clearing time, δ_1 is post-fault stable state angle, and δ_{max} is the maximum allowable angle at post fault equilibrium point. Prior to a fault, the VSG operates at point *a*, and the power angle is δ_0. However, when the fault arises, the operating point deviates suddenly from point *a* to point *b*. At point *b*, VSG remains in acceleration mode because the input power of VSG is higher than the output power. Meanwhile, load angle increases and operating point moves from point *b* to point *c*. Furthermore, when the fault clears successfully at point *c*, and operating point jumps from point *c* to point *e*. At point *e*, the input power of SG is less than the output power, and VSG enters in the deceleration stage. During deceleration, the virtual angular frequency decreases, and the power angle continues to increase until the virtual angular velocity at point *g* is equal to rated angular frequency.

However, if the system goes below the rated angular frequency and load angle continue to increase and passes the maximum allowable rotor angle limit, then the system enters into acceleration region again and loses synchronism. Therefore, it is necessary to keep the system beyond the maximum allowable angle. The characteristics of SG and conventional VSG are similar enough, however conventional VSG possess few characteristic and control structure differences due to which the power-angle curve of conventional VSG is somewhat different from SG. Moreover, during the large transient faults, conventional VSG lost synchronism and became unstable and causes the whole system to become unstable.

Mathematically, the acceleration area A and deceleration area B can be stated as follows:

$$\begin{aligned} A &= \int_{\delta_0}^{\delta_c} (P_m - P_3)d\delta \\ B &= \int_{\delta_c}^{\delta_1} (P_2 - P_m)d\delta \end{aligned} \tag{6}$$

According to Fig. 3, if acceleration area A is less then deceleration area B, then the power system considers as stable. If acceleration area A is greater than deceleration area B, then the power system is unstable. However, if area A and area B are equal, then the power system is marginally stable. Consequently, the transient stability of the system can be improved by reducing the acceleration area or increasing the deceleration area in the transient process. Moreover, this criterion is consistent to the transient energy function; therefore, it can be used for two machine power system consisting of VSG and SG. Therefore, for a pair of machines, the rotor swing equation can be written as follows:

$$\begin{aligned} \frac{d\omega_{12}}{dt}\left(\frac{M_1M_2}{M_1+M_2}\right) &= \left(\frac{P_{m1}M_2}{M_1+M_2} - \frac{P_{m2}M_1}{M_1+M_2}\right) \\ &- \left(\frac{M_2(P_{e1}+P_{e12})}{M_1+M_2} - \frac{M_1(P_{e2}+P_{e21})}{M_1+M_2}\right) \end{aligned} \tag{7}$$

The acceleration area A and deceleration area B for two machines can be stated as follows:

$$\begin{aligned} A &= \int_{\delta_{0,12}}^{\delta_{c,12}} (P_{m,12} - P_{3,12})d\delta \\ B &= \int_{\delta_{c,12}}^{\delta_{1,12}} (P_{2,12} - P_{m,12})d\delta \end{aligned} \tag{8}$$

4. Proposed modification in VSG control

In order to analyse the transient stability of the VSG based power system, the equal-area criterion is used, which is based on power-angle characteristics of the system. Moreover, VSG input power can be affected by various factors such as variable output from renewable energy sources and their limited controllability. Therefore, the auxiliary inertia and additional power are incorporated in the active power-frequency control loop of VSG.

4.1. Auxiliary inertia

The auxiliary inertia incorporation in governor unit is illustrated in Fig. 4. The governor unit of conventional VSG is somewhat different and regressive than the governor unit of SG because of the absence of inertia in it. Due to this difference, the power and frequency stability response of conventional VSG is quite distorted in comparison with SG during significant fault or disturbances.

Mathematically the simple governor control of conventional VSG, including droop control, can be stated as:

$$\begin{aligned} T_m &= \frac{P_m}{\omega_r} \\ P_m &= P_{ref} - k_p(\omega - \omega_r) \end{aligned} \tag{9}$$

where k_p is the angular frequency regulation coefficient. To compete with the governor unit of the SG, auxiliary inertia is added into the conventional VSG governor unit and mathematically it can be represented as:

$$T_m = \frac{k_a(\omega_r - \omega + k_e(P_m - (P_e + P_\delta)))}{(s + k_c)s} \tag{10}$$

where k_c is the time constant, k_a is the gain, k_e is the angular frequency

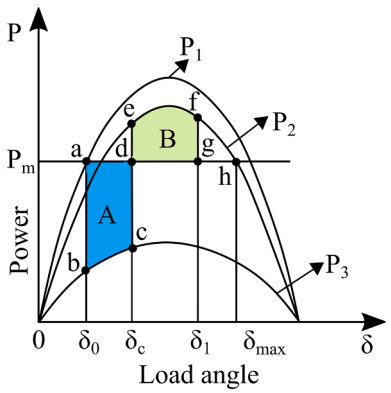


Fig. 3. Power-angle characteristic curve.

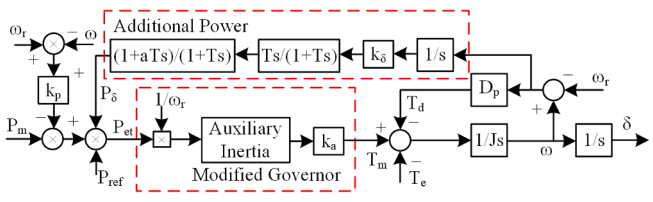


Fig. 4. Control diagram of VSG with additional power and auxiliary inertia.

coefficient, and P_δ is the auxiliary power. This auxiliary inertia improves the stability and convergence time of new Modified-VSG (M-VSG) in comparison to the conventional VSG. Moreover, auxiliary inertia helps to improve the power and frequency stability during system faults, and M-VSG governor unit behaves as governor unit of SG. Consequently, M-VSG consist of two different separate inertia, therefore the flexibility of VSG widens in comparison to conventional VSG.

4.2. Additional power

In order to improve the power-angle stability of the system by applying the equal area criterion, it is necessary to increase the active output power of VSG. Therefore, to fulfil this task, additional power is incorporated into the active power-frequency loop as depicted in Fig. 4. According to this method, during the large transient fault, a grounding resistance is connected parallel to the VSG, which increase the output active power and prevent the VSG from losing stability. Therefore, the addition of auxiliary power into power-frequency loop changes the swing equation, which can be rewritten as:

$$\begin{aligned} J\frac{d\omega}{dt} &= T_m - T_e - D(\omega - \omega_r) \\ \frac{d\delta}{dt} &= \omega \\ P_\delta &= k_\delta \Delta\delta \frac{Ts(1+\alpha Ts)}{(1+Ts)^2} \end{aligned} \tag{11}$$

where k_δ is the synchronisation coefficient, α is the lead-lag component which needs to compensate for the phase delay of the active power-frequency loop of VSG and T is the time constant of the isolation link, which is used to avoid static error interference during steady state operation. Therefore, the power-angle curve of M-VSG which possess auxiliary inertia and additional power is depicted in Fig. 5. Moreover, the acceleration area and deceleration area terms can be modified as follows: (see Fig. 6)

$$\begin{aligned} A_1 &= \int_{\delta_0}^{\delta_c} (P_m - P_3)d\delta - \int_{\delta_0}^{\delta_c} P_{aux}d\delta \\ B_1 + B_2 &= \int_{\delta_c}^{\delta_1} (P_1 - P_m)d\delta + \int_{\delta_c}^{\delta_1} P_{aux}d\delta \end{aligned} \tag{12}$$

where, P_{aux} is the power curve of VSG after incorporating the additional power. Fig. 5 depicts the power-angle curve of M-VSG. This curve shows that the acceleration area of M-VSG is smaller than the deceleration, which confirms that the M-VSG enhanced the stability of the power system. Moreover, auxiliary inertia and additional power reduce the amplitude of frequency oscillation and suppress the frequency exceeding the threshold in the transient process.

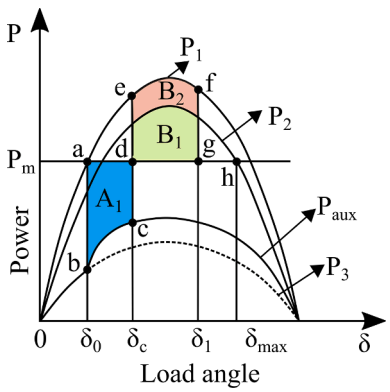


Fig. 5. Power-angle characteristic curve of M-VSG.

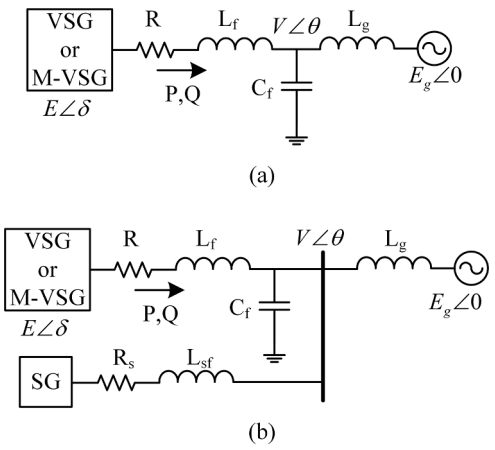


Fig. 6. Single line diagram (a) Case I (b) Case II.

5. Simulation results

In order to verify the effectiveness and validate the proposed modification, the simulation model is developed according to the electric circuit shown in Fig. 5(a) and (b). Two cases are assumed as the case I and case II for Fig. 5(a) and (b), respectively. The short circuit fault occurs at time 2 s and clear after 0.1 s for both cases. The detailed system parameters and control parameters are listed in Table 1 and Table 2.

5.1. Case I

In this case, the conventional VSG and M-VSG are connected to a large grid separately. The comparative analysis of conventional VSG and M-VSG is shown in the following figures.

Fig. 7 presents the dynamic responses of the conventional VSG and M-VSG for a short circuit fault which occurs at 2 s and eliminates at 2.1 s. It can be observed that when conventional VSG is connected to the grid and fault occurs, it starts to deviate, and this deviation increases continuously. In fact, when the fault occurs, conventional VSG enters into acceleration area, and δ starts to deviates. However, when fault eliminated at δ_c from the system, conventional VSG enter into deceleration area, and δ continue to increase; eventually, it crosses the δ_{max} limit and enters into acceleration area again, and the system became unstable as depicted in Fig. 8(a).

Though, when M-VSG is connected to grid and fault occurs, it goes into acceleration area, and δ starts to increase. As soon as fault clear at δ_c, it enters into deceleration and goes up to δ_1. However, the auxiliary inertia in governor block and additional power term opposes the further increment in δ, and then it turns back as illustrated in Fig. 8(b). Moreover, in Fig. 7, it can be seen that the power-angle deviation of M-VSG is quite insignificant, and the power-angle stabilises soon after the fault is cleared.

The frequency response is depicted in Fig. 9, and it can be seen that when the fault occurs and conventional VSG is connected with grid, frequency deviates up to 50.7 Hz. Frequency continues to oscillate even

Table 1
VSG parameters.

Parameters	Values	Parameters	Values
Grid Voltage	400 V	C_f	0.01
Rated power	20 kVA	L_g	0.33
Fundamental Frequency	50 Hz	D_p	35
Auxiliary Inertia	.5	k_q	3.18
Inertia	.95	k_a	1.59
L_f	0.02	k_δ	3.1
R	0.037	k_p	2.6

Table 2
SG parameters.

Parameters	Values
Rated Power	18 kVA
Rated Voltage	13.8 kV
Inertia	1.4
Rated pf	0.85

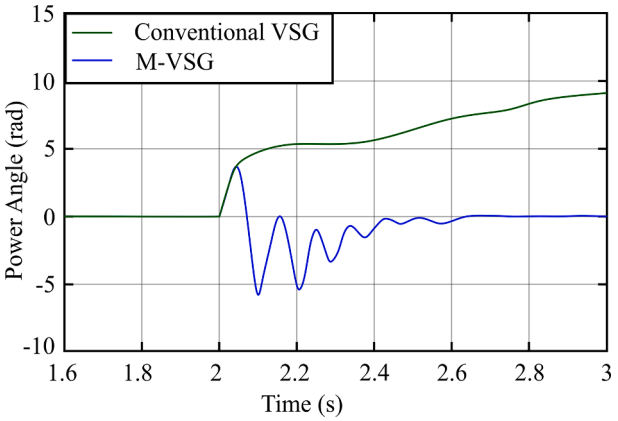


Fig. 7. Power-angle response of conventional VSG and M-VSG for case I.

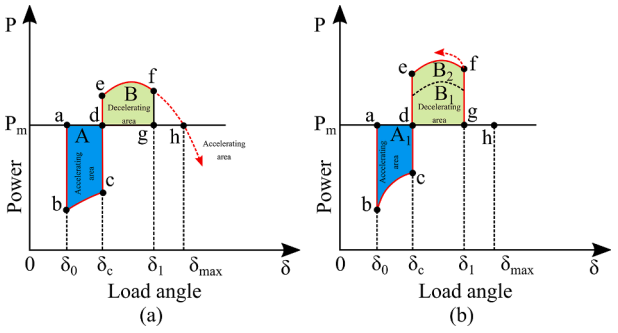


Fig. 8. Power-angle characteristic curve for whole transition process of (a) conventional VSG (b) M-VSG.

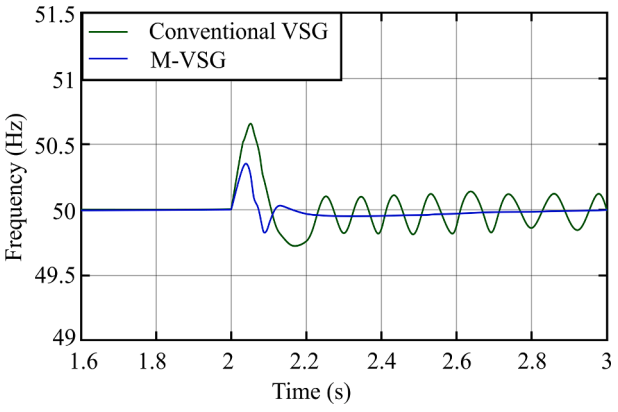


Fig. 9. Frequency response of conventional VSG and M-VSG for case I.

fault is removed from the system. However, when M-VSG is connected to the grid and fault occurs, the frequency star to deviate, but auxiliary inertia of governor unit opposes the significant frequency deviation. It is observed that frequency deviation is quite low in comparison to conventional VSG frequency response. Moreover, M-VSG frequency achieves steady state quickly after the fault clear from the system.

In Fig. 10, the active power response of VSG and M-VSG is illustrated. It is depicted in output that the active power of conventional VSG deviates significantly during fault and start to fluctuate after the fault clearly from the system. However, the auxiliary inertia and additional power in M-VSG oppose the significant deviation in the active power during the fault. Once the fault clear from the system, M-VSG oscillates slightly and then gains the so-called steady state.

Fig. 11 shows the reactive power response for conventional VSG and M-VSG. The reactive power deviates significantly and starts oscillation during and after the fault when conventional VSG is integrated with the grid. The reactive power response of M-VSG is slightly different from conventional VSG during grid integration. The reactive power deviates during fault significantly, however, achieve steady state upon fault clearance after some oscillations.

Fig. 12 shows the total power for conventional VSG and M-VSG. It is observed that the conventional VSG power curve is in a steady state before fault, whereas, during and after the fault, the conventional VSG power curve deviates and starts oscillation. However, the power curve of M-VSG deviates during the fault, but it quickly converges its steady state after fault clearance.

5.2. Discussion

It is observed that when conventional VSG and M-VSG connected to a large grid, and a significant fault occurs then both VSGs behaves quite differently. Considering the equal area criterion, the power-angle of conventional VSG suffers significant deviations. The conventional VSG enter into acceleration region during fault and goes into deceleration area as soon as fault clears; however, the δ continue to increase, and it crosses the δ_{max} and goes into the acceleration area. When it passes the δ_{max}, the acceleration area becomes large, and deceleration area became small, which shows that system enter into the unstable region and loss synchronism as shown in Fig. 8(a).

On the other hand, M-VSG response during and after significant fault is quite different and reasonable. During a fault, M-VSG goes into acceleration area and turn toward the deceleration area upon fault clearance, and δ continue to increase. However, the additional power term and auxiliary inertia of governor unit keep the δ beyond the maximum allowable limit, reduces the acceleration area and increases the maximum possible deceleration area; therefore system remains in the stable state as shown in Fig. 8(b).

5.3. Case II

For this case, the effects of conventional VSG and M-VSG integration parallel to SG are observed dynamic state. The diverse characteristics for power regulation of SG and VSG cause massive power fluctuation when

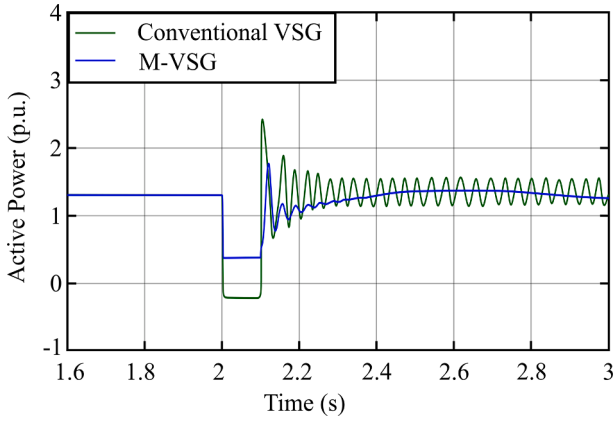


Fig. 10. Active power response of conventional VSG and M-VSG for case I.

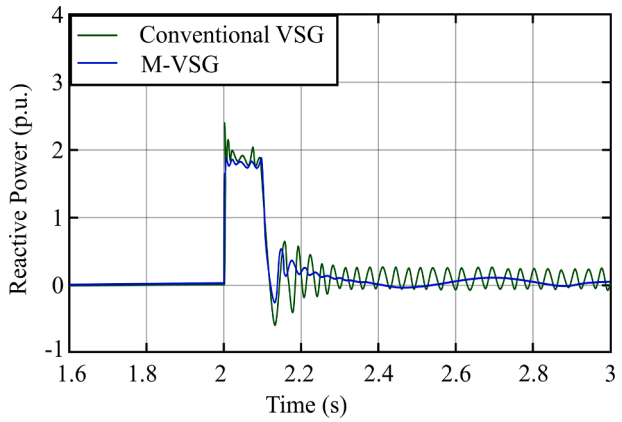


Fig. 11. Reactive power response of conventional VSG and M-VSG for case I.

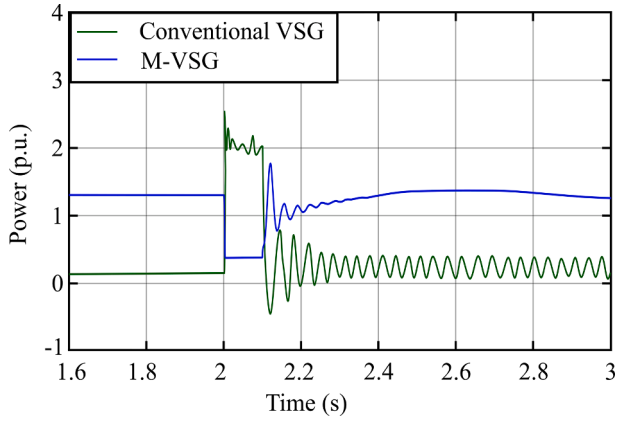


Fig. 12. Total power response of conventional VSG and M-VSG for case I.

both generators are connected in parallel. Therefore, the auxiliary inertia of governor unit helps to suppress the fluctuations caused by disturbances during parallel integration. The comparative analysis results of conventional VSG, M-VSG and SG are shown in the following figures.

The power-angle response is shown in Fig. 13, and it can be observed that when conventional VSG is connected parallel to the SG and fault occurs, the power-angle deviates significantly and continue to deviate even after fault is cleared. This deviation of power-angle is larger in comparison when conventional VSG is connected to a large grid alone, as depicted in Fig. 7. However, M-VSG acts as an SG effectively. When M-VSG is connected parallel to SG, the power-angle deviates slightly and stabilises quickly due to auxiliary inertia of governor unit and additional power.

The frequency response for parallel integration of conventional VSG, M-VSG with SG is shown in Fig. 14. It can be observed that when the fault occurs, the conventional VSG experience large frequency deviation in comparison case I, as illustrated in Fig. 9. However, when M-VSG is connected parallel to the SG and fault occurs, the frequency deviates slightly due to the auxiliary inertia and additional power terms. The M-VSG governor unit significantly improves the frequency response for parallel integration scenarios. Moreover, frequency achieves steady state quickly after fault eliminated from the system.

In Fig. 15, the active power response is illustrated. It can be seen that during the fault when conventional VSG integrated parallel to SG the active power of conventional VSG deviates significantly in comparison to case I Fig. 10 and continue to fluctuate even the fault is cleared from the system. However, when M-VSG is integrated parallel to SG, the active power of M-VSG deviate slightly during the fault due to the auxiliary inertia and additional power in M-VSG opposes the significant deviation. Consequently, after fault eliminated from the system, active power oscillates slightly in comparison to the active power of SG.

The reactive power response of parallel integration conventional VSG and M-VSG with SG is shown in Fig. 16. It is illustrated in Fig. 16 that during parallel integration of conventional VSG and SG, the reactive power deviation of conventional VSG is large enough. When the fault eliminates from the system, the reactive power of conventional VG continues to oscillate. Whereas, when M-VSG is integrated parallel to SG, the reactive power deviate slightly during fault but achieve a somewhat steady state upon fault elimination after slight oscillations.

In Fig. 17, the total power of SG, conventional VSG and M-VSG are shown. It is shown that during and after the fault, the conventional VSG power deviates and starts oscillation. The conventional VSG experience significant power fluctuation due to parallel integration with SG. However, the power of M-VSG deviates during the fault, but it quickly gains the steady state after fault clearance. Moreover, the power response of parallel integrated M-VSG and SG is fairly similar during and after the fault because of the auxiliary inertia of governor unit and additional power.

5.4. Discussion

For case II, it can be observed that when SG in parallel to conventional VSG or M-VSG connected to a large grid, and a significant fault occurs, then conventional VSG behaves quite differently. In contrast, M-VSG behaviour is somewhat similar to the SG. According to the equal area criterion, the power-angle of conventional VSG suffers significant deviations in comparison to SG. Whereas the power-angle of M-VSG is quite similar in this case. Moreover, it is shown in results that the frequency response, active and reactive power response of conventional

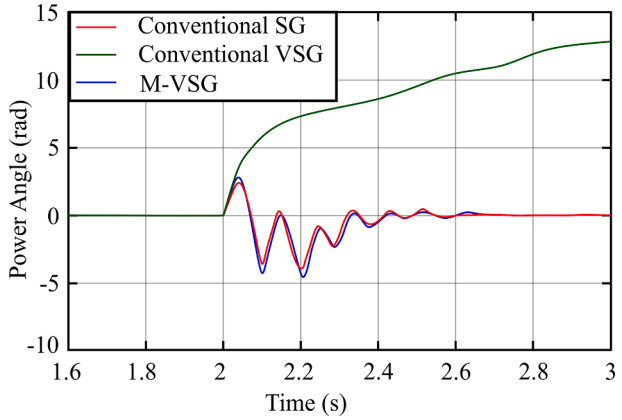


Fig. 13. Power-angle response of conventional VSG, M-VSG and SG for case II.

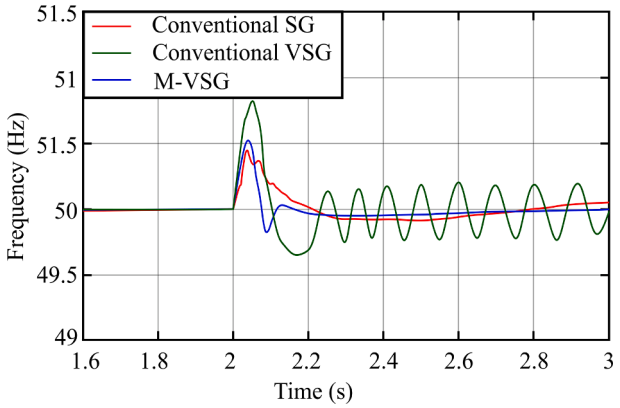


Fig. 14. Frequency response of conventional VSG, M-VSG and SG for case II.

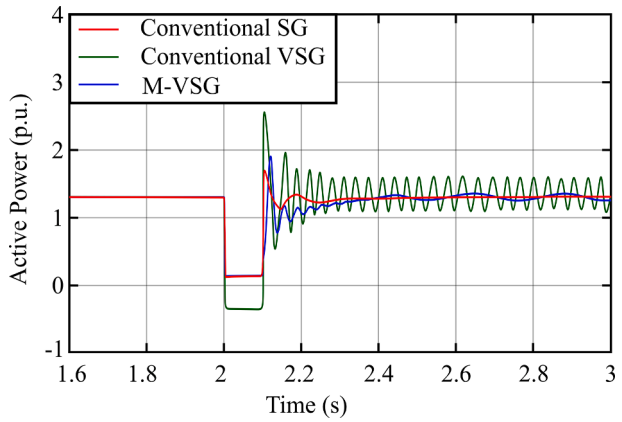


Fig. 15. Active power response of conventional VSG, M-VSG and SG for case II.

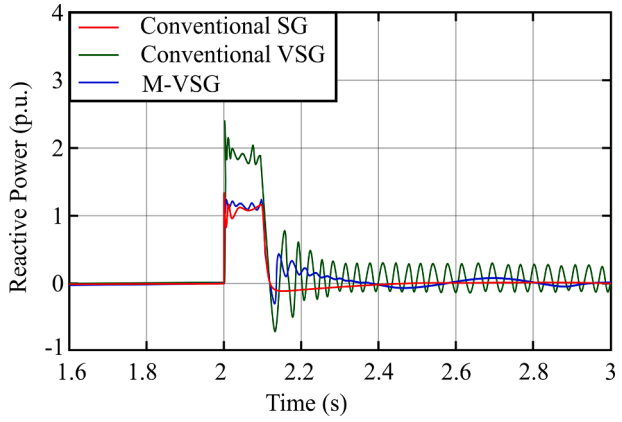


Fig. 16. Reactive power response of conventional VSG, M-VSG and SG for case II.

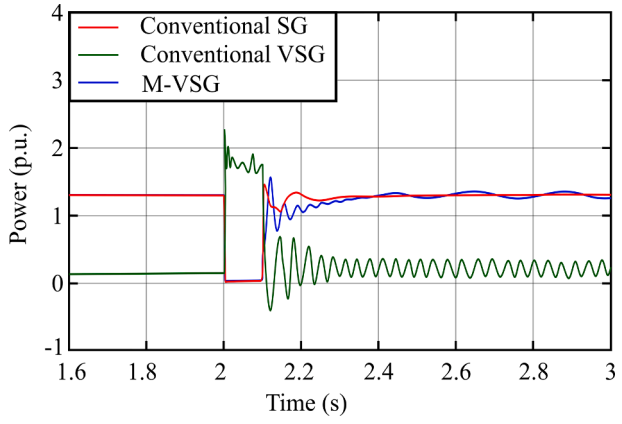


Fig. 17. Total power response of conventional VSG, M-VSG and SG for case II.

VSG, is quite distorted, while M-VSG follows the SG effectively.

5.5. Case III

The sudden change in load causes severe fluctuations in the system. Therefore, in this case, the dynamic response of conventional VSG and M-VSG is observed and analysed. The sudden load change occurs on the system at 2sec and loads value change from 3 kW, 1 kvar to 10 kW, 3 kvar, when the conventional VSG and M-VSG are connected parallel to SG, separately. The VSGs and SG capacities are almost similar; therefore, the power-sharing to the load is according to the VSGs and SG capacity. Additionally, the power and frequency variations of conventional VSG are huge when integrated parallel to SG. The comparison of simulation results of conventional VSG, M-VSG and SG is presented in the following figures.

The power angle response is shown in Fig. 18; it can be seen that power angle deviate significantly and losses synchronism and VSG enter into an unstable state. When load change suddenly on the network, and conventional VSG is connected parallel to SG, rotor angle of conventional VSG loses synchronism. In contrast, the power angle response of M-VSG is quite different from the power angle response of conventional VSG. The power angle of M-VSG is relatively similar to the response of SG due to the auxiliary inertia and additional power loop. When the load fluctuates, the power angle of M-VSG changes accordingly. However, the proposed modification in M-VSG control allows it to behave quite similar to the SG.

The frequency response of conventional VSG, M-VSG with SG for abrupt load change is shown in Fig. 19. When the load increase suddenly, the frequency of conventional VSG deviate significantly and then continues to oscillate because conventional VSG goes into the unstable region. Whereas the frequency of M-VSG deviate relatively at a moderate level in comparison to conventional VSG and it acts as somewhat similar to SG. Over the time, M-VSG frequency goes into relative stabilisation as SG, and it can be observed in Fig. 19.

The active power response for case III is shown in Fig. 20. I can be observed in Fig. 20, that when the sudden load came onto the system, the active power of conventional VSG shoots up and downwards and continue to oscillate. On the other side, the active power response of M-VSG is relatively different. When the load came into system suddenly, the active power of M-VSG fluctuate; however, it gains steady state quickly due to the auxiliary inertia and addition power loop in M-VSG, which makes a VSG to perform functions as an SG.

In Fig. 21, the reactive power response of parallel integration, conventional VSG and M-VSG with SG are shown, due to sudden load change. It can be seen that the reactive power of conventional VSG deviates as soon as load change; however, the reactive power of conventional VSG could not achieve a steady state due to a weak control algorithm. In contrast, the M-VSG start delivering the reactive power to the load. The reactive power of M-VSG gains steady state quickly without losing synchronism due to its modified and improved control algorithm.

5.6. Discussion

For case III, the conventional VSG and M-VSG are connected parallel to the SG separately, and the dynamic response of VSGs and SG is observed when a large load is applied to the system. The above figures and their analysis show that when conventional VSG is integrated

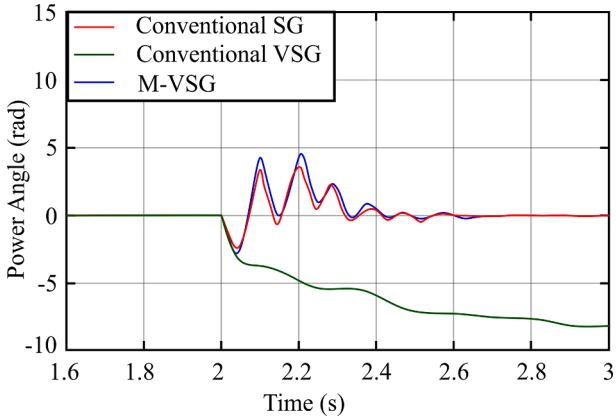


Fig. 18. Power-angle response of conventional VSG, M-VSG and SG for case III.

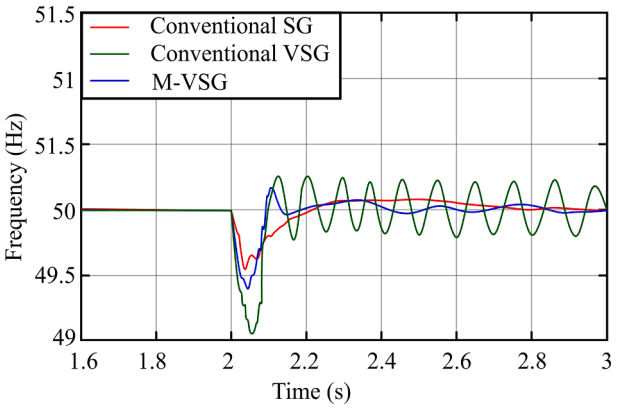


Fig. 19. Frequency response of conventional VSG, M-VSG and SG for case III.

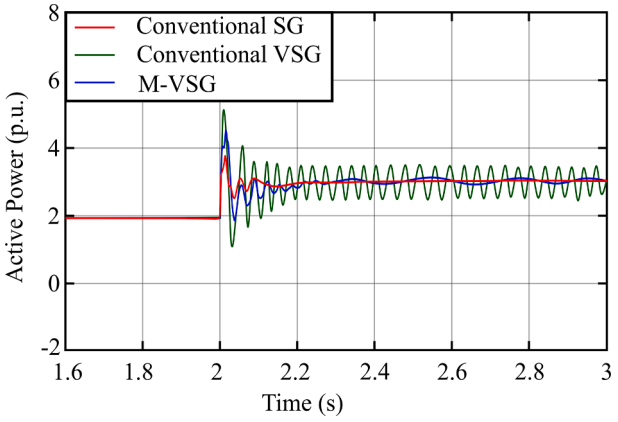


Fig. 20. Active power response of conventional VSG, M-VSG and SG for case III.

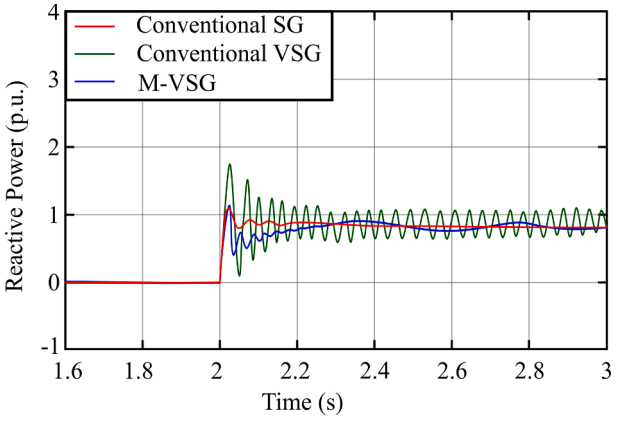


Fig. 21. Reactive power response of conventional VSG, M-VSG and SG for case III.

parallel to SG and load change suddenly, it loses synchronism with the system and moves into an unstable region according to equal are criterion. The parallel integration of conventional VSG and SG is vulnerable to large fluctuation and irregular behaviour because of various differences in the control structure. Therefore, as soon as the load applied to the system, conventional VSG loses synchronism.

In contrast, the M-VSG performance is quite different from conventional VSG for the very same parallel integration with SG. The above analysis shows that the M-VSG behaves quite similar to the SG. When the load came to the system abruptly, M-VSG loses it steady state for a short time; however, the proposed auxiliary inertia and additional power loop control helps M-VSG to regain its steady state equilibrium at a different position.

Additionally, various techniques are proposed to improve transient stability of VSG, such as incorporating auxiliary damping control into VSG control, apply centre of inertia technique to the VSG, integrating the improved power decoupling technique to VSG control. Mostly, these techniques are suitable for islanded or multiple VSG system and significantly difficult to implement and analyse. However, the incorporation of additional power and auxiliary inertia into VSG control and the stability analysis with equal area criterion is relatively easy for single or two machines connected to a large grid in comparison to other techniques.

6. Conclusion

In this paper, the power angle stability problem for VSG during the transient process under significant system fault is addressed. In order to suppress the power angle instability of VSG, an additional power term long with additional inertia added in active power loop of the VSG. The additional power reduces the acceleration area and increases the maximum possible deceleration area in the transient process, while auxiliary inertia provides better frequency support capability for the power system. Moreover, theoretical analysis and simulation results show that the proposed control strategy effectively reduces the fluctuation of M-VSG power angle, frequency and active and reactive power when M-VSG connected to grid or parallel to the SG. Furthermore, the convergence time of the proposed strategy is significantly better than conventional VSG strategy due to. However, to improve the proposed strategy further, the VSG flexibility and parameter adjustment require a detailed analysis.

CRediT authorship contribution statement

Khalid Mehmood Cheema: Conceptualization, Methodology, Software, Writing - original draft. **Ahmad H. Milyani:** Visualization, Validation, Writing - review & editing, Funding acquisition. **Ahmed M. El-Sherbeeny:** Visualization, Validation, Writing - review & editing, Funding acquisition. **Mohammad A. El-Meligy:** Visualization, Validation, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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